

A pre-feasibility study for Producing Organic Fertilizers Project In Ajloun Governorate



2017



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1. Executive Summary

The project aims to establish a specialized plant in the production of organic fertilizers. The project is based on the production of organic fertilizers manufactured from agricultural wastes by obtaining the solid residues resulting from plantations, animal dung and poultry dung for fermentation, drying, sterilization and packaging to produce organic fertilizers. In accordance with the instructions No. G / 6 of 2016 of the terms and conditions of licensing the production of fertilizers, cultivation regulations, processing, storage, circulation, trade and advertising. As well as Jordanian Standard 962-2011 of organic fertilizers and soil fertilizers. So that they are free of pathogens and seeds of plants and adopt the technical specification of the compost mentioned in the instructions, and the package capacity will be of 25 kg.

Table 1: Economic Indicators

Project Name	Organic fertilizers Production
The Proposed Site	Ajloun
The Products of the Project	Organic Fertilizers
Total Labor	8
Total Investment (,000 JD)	258
Net Present Value of Cash Flows (,000 JD)	383
Payback Period (Years)	5
Internal Rate of Return	%25.59

The results of the financial analysis indicate that the net value of the cash flows of the project is positive, which is an indicative of the financial feasibility of the project and the internal rate of return is much higher than the weighted average cost of capital. Other financial indicators indicate that the project is profitable to the investor.

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

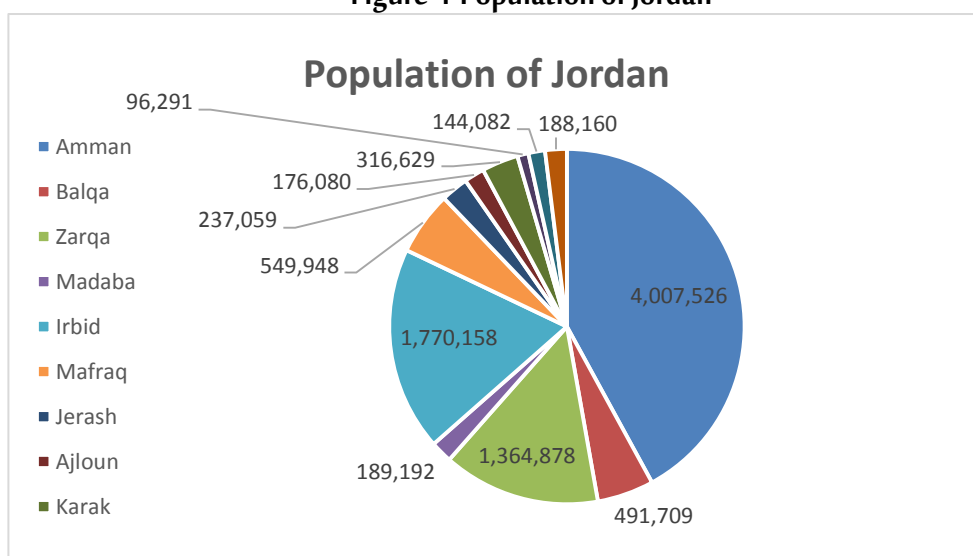
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 1: Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

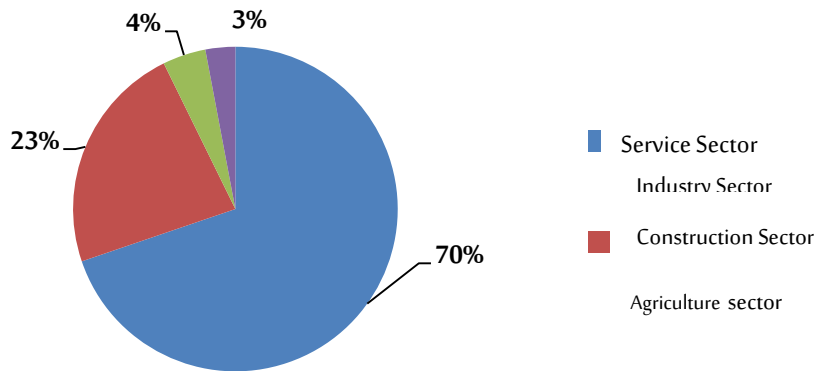
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is

dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce. Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market Analysis

Project Description

Organic fertilizers are all that is added to agricultural soil to supply organic matter and humus to provide developing plants with minor and major nutrients.

The project is based on the idea of covering the increasing demand for organic fertilizers by producing high quality of organic fertilizers free from viral, fungal and bacterial pathogens based on their production through aerial fermentation processes in order to obtain high quality products through which to preserve and improve natural properties such as soil To maintain fertility and improve the properties of water retention and to provide the plant with the necessary nutrients.

The project is one of the industrial and agricultura service projects I aims to produce various fermented organic fertilizers (compost) used in agriculture and to supply the local market with many types of decaying organic fertilizers free of pathogens and seeds of exotic plants. The compost is used by farmers in large quantities to help the growth o crop and improve their quality .It is known that compost or organic fertilizer is preferable than the types made of chemicals such as urea and others.

Project Objectives

Due to the high demand for quantities of organic fertilizers yearly, as a result of the development of irrigated cultivated area, which need increasing amounts of fertilizer as a result of soil stress, the project aims to produce organic fertilizers from the remnants of agricultural crops, animal dung and poultry glands in Ajloun Governorate.

The project aims to:

- 1- Achieving a profitable return to the owner of the project,
- 2- Production of good quality organic fertilizers and supply them to the local market,
- 3- Export local production of organic fertilizer to neighboring markets,
- 4- Creating Job opportunities in the less developed regions,
- 5- Exploitation of agricultural wastes and conservation of the environment.

Proposed Product

It is advisable to avoid land erosion by excessive use of chemical fertilizers. If these chemical fertilizers actually provide the plants with the essential elements they need of , nitrogen, potassium, phosphorus, sulfur, iron and manganese, but they will not promote the growth of bacteria which will contribute in the fragmentation of organic matter in the soil and converted to metal materials, that are supplied by chemical fertilizers only in extra quantities, in contrast to organic fertilizer. As organic fertilizers produce these bacteria intensively, thus depleting the soil year after year with mineral materials:thus the proceeds of plantations grow permanently. Organic fertilizers will be produced by the use of the best modern technology and scientific means to ensure the quality of the product.

It was discovered that agricultural residues has a high economic return, investors began to work on the transformation of those waste to a material of economic value is competing for how to exploit it, whereas those wastes are considered in their accumulation a large burden on the farm and the farmer and unhealthy phenomenon on the environment .currently attention is beyond the agricultural production limit and how to increase the amount of the crop to achieve the highest possible productivity, but al it extends to find a very comprehensive approach to ensure the utilization of agricultural residues generated from those crops to reuse and replace the common health concept that recognize them as a great burden on the farms and the environment in their accumulation and burn them ,by the correct concept of economic value as investors are competing in how to re-use, and the diversity of investments.

Organic material plays an important role in solving soil problems where it is an important part of a complex absorption that retains nutrients. It is easily accessible to plants. Due to the increase in the reciprocal capacity of organic material alkaline, compared to clay minerals, it leads to decreased in nutrient loss rates. At the same time, organic material is an important source of energy for most soil organisms, and the decomposition of organic material is unique in its components of nutrients, from which plants benefit from it. The organic acids also help to increase the use of some non-plant-friendly elements like phosphorus and iron. It was found that the addition of organic fertilizers to the sandy soil leads to increase land coherence, fold quickly leakage of irrigation water property, increase its capacity to retain water, increase soil fertility and provide crops by nutrient needs, also the organic material has a major role in soil aeration and cohesive clay, overcoming the problems of calcareous soils and alkaline where added to improve natural characteristics.

The sources of organic fertilizers are livestock and poultry manure, which is a natural fertilizer that helps to improve the properties of agricultural soil and is one of the oldest types of fertilizers known to humans, but the wet state makes it difficult to handle as well as transferring long distances, and being in this situation has a negative impact on the environment, Hence, the idea of establishing the proposed project for the production and packing of natural fertilizers in packages suitable for long distance transport and supply to farmers and garden owners in a manner that does not harm the environment.

Market Size Analysis

Organic fertilizer market in Jordan is a huge market as a result of the expansion of irrigated crops, as all crops use different amounts of organic fertilizers due to their different characteristics that distinguish them from chemical fertilizers. Organic fertilization is the cornerstone that should be developed to raise the productive value of agricultural land and reduce the environmental pollution resulting Of the waste in the use of mineral-chemical fertilizers and therefore recycling organic waste is one of the important factors that lead to the provision of quantities of organic fertilizers that meet the needs of agricultural land.

The data of the Department of Statistics indicate that the production of organic fertilizer from animal residues (cows, sheep, goats and chickens reached 452 thousand tons in 2015) (table 4), however, there is a significant amount of manure that have not to been exploited and disposed of either in landfill or burned. The environmental statistics by the Department of Statistics indicate that there were about 68 thousand tons of animal residues have been disposed of in different ways and about 57 thousand tons left in the open air , which is an environmental scourge. There is a great opportunity to exploit these wastes (Table 2)

Table 2 : Evolution of the Kingdom's production of untreated organic fertilizers

Year	Chicken		Total Livestock	
	Quantity (Tons)	Value (JD)	Quantity (Tons)	Value (JD)
2011	116,360	1,012,477	277,239	2,576,449
2012	140,223	1,229,802	259,696	2,280,090
2013	103,623	966,950	333,014	2,972,412
2014	117,616	1,219,936	392,669	3,449,221
2015	96,753	958,236	452,324	3,926,030
AVG	114,915	1,077,480	342,988	3,040,840

Source: Department of statistics

Table 3 : Quantity of Livestock Residues by Region and Disposal Methods for 2011

Region	Type of possession	Disposal Method				Total Quantity (Tons)
		Garbage	Burn	Throw away	Other	
North	Rigestered Cows	0	0	697	835	1,532
	Unregistered Cows	105	0	172	0	277
	Sheep and Goats	49	1	20,147	204	20,401
	Chicken	0	45	492	0	537
	Total	154	46	21,508	1,039	22,746
Middle	Rigestered Cows	0	5,131	134	31	5,296
	Unregistered Cows	189	460	1,824	232	2,705
	Sheep and Goats	36	643	19,522	28	20,229
	Chicken	82	810	298	0	1,190
	Total	307	7,045	21,778	291	29,420
South	Rigestered Cows	0	0	33	0	33
	Unregistered Cows	0	0	30	8	37
	Sheep and Goats	665	72	12,527	12	13,277
	Chicken	1,014	67	1,134	0	2,215
	Total	1,680	139	13,724	20	15,563
Total		2,141	7,229	57,010	1,349	67,729

As there is no accurate data and information about the actual demand for organic fertilizers or chemical in Jordan or about the fertilizer volume needed for one dunum of cultivated land, whereas the needs vary depending on soil type, according to cultivated varieties and available irrigation method, some farmers estimate these needs between 0.5 - 3.0 tons per dunum for the irrigated cultivation, while these requirements is estimated at about 250-500 kg per dunum of rain-fed land, on the basis of the minimum, the total requirement of fertilizer for the areas mentioned above is estimated at about 650 thousand tons per year, it is obtained mostly raw form of poultry and livestock farms in violation of the environment law, also prevents the supply of fermented organic fertilizers to the lands of Jordan Valley.

The demand for organic fertilizers is derived from agricultural areas. The irrigated area in Jordan amounted to about 1 million dunums. Table (4) shows the evolution of irrigated area cultivated by fruit trees, vegetables and field crops, and we have assumed conservative figures of the organic fertilizers required for the cultivation of one dunum (1 ton for vegetables and half tons for trees as organic fertilizer is added every two years and a half ton for field crops and rainfed crops are neglected) Estimation of initial demand for organic fertilizers during the coming period. Table (5) shows the quantities of demand during the period preceding the organic fertilizer, based on the previous assumptions.

Table 4: Development of Irrigated Area in Jordan in Donum

Irrigated Area	2011	2012	2013	2014	2015
Fruit trees	469,751	472,412	450,947	457,631	476,544
Field Crops	87,549	84,109	111,133	114,405	96,825
Vegetables	407,195	400,018	472,353	478,436	461,433
Total Irrigated Area	964,495	956,538	1,034,433	1,050,471	1,034,801

Source: Department of Statistics

Table 5: Evolution of demand for organic fertilizers based on irrigated areas

Irrigated Area	2011	2012	2013	2014	2015
Fruit Trees Demand	234,875	236,206	225,473	228,816	238,272
Field Crops Demand	43,774	42,054	55,567	57,202	48,412
Vegetables Demand	407,195	400,018	472,353	478,436	461,433
Total Demand	685,845	678,278	753,393	764,454	748,117

The data of EU exports of organic fertilizers to Jordan, show annual exports ranging between 100 to 2000 tons. (Table 6) and most of the fertilizers were imported from Italy. Table (7) shows quantities of imports, exports, local production and total demand for organic fertilizer during the period (2011-2015).

Table 6: EU exports of organic fertilizer to Jordan

Year	Exports Value (Thousand EUR)	Exports Quantity (Tons)	Price of tons of exports (EUR / ton)
2011	874	2,169	403
2012	756	1,940	389
2013	568	880	646
2014	351	710	494
2015	176	109	1,611

The demand for organic fertilizers was estimated in two ways that led to the same result. The first method based on the general trend of local production of organic fertilizers, And the second by calculating the rate of consumption per capita, where the quantities of previous demand were divided by the population to obtain the equivalent per capita. As the consumption rate is linked to the population as a result of the increase in demand for foodstuffs, and since the population growth rate in the Kingdom according to the data of the Department of Statistics is 2.2% annually, assuming that the growth rate of demand is equal to the population growth rate.

Table7 : Evolution of imports and domestic production of organic fertilizers during the period (2011-2017)

The size of the Jordanian market of organic fertilizers	2011	2012	2013	2014	2015	2016	2017*
Imports (tons)	2,169	1,940	880	710	109	111	113
Exports (tons)	390	742	160	0	0	0	0
Re-exported (tons)	0	0	1,7	0	9	9	9
Domestic production of the organic fertilizers (ton) (2)	277,239	259,696	333,014	392,669	452,324	461,370	470,598
Total demand for organic fertilizers (tons)(2)	685,845	678,278	753,393	764,454	748,117	763,079	778,341

Source: Department of Statistics (2015), (2) Central Bank (2015)

Table 8: Estimated demand on organic fertilizers during the period (2018-2027)

The quantity of the demand	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Expected population (1,000 people)	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144	10,204
Average per capita annual consumption (kg / person)	0.078	0.078	0.078	0.078	0.078	0.078	0.078	0.078	0.078	0.078
Demand derived on organic fertilizers (tons)	795,464	812,965	830,850	849,129	867,809	886,901	906,413	926,354	946,734	795,464

Competitors Analysis

There are three main factories in the Kingdom that Process and treat the waste of poultry and livestock, where fermenting and drying processes are done, and the addition of pesticides to kill the larvae and seeds of the suspended herbs and then grind and fill them in bags. These factories are:

- Jordan International Industries Company L.P.C/ Halabat, estimated production capacity of about 35000 tons per year.
- Jordan Agricultural Company for the manufacture of organic fertilizers / Al-Halabat, and its production capacity is estimated at 12000-15000 tons per year.
- The Arab Company for the manufacture of Organic Fertilizers / Knaker, and its production capacity is estimated at 6000 tons per year.

In addition to the above projects, there are several other small projects that are drying, packing and marketing natural fertilizers, but the quality of production is not good. There are complaints from a number of farmers and nurseries that these fertilizers are not well treated and contain insects and plant seeds, Resulting in their farms and nurseries.

Therefore, the project is not expected to face difficulties in achieving the expected sales if it targets the local market only at competitive prices. Therefore, the investor must develop a good marketing plan and employ competencies that have the ability to communicate with senior farmers and direct marketing instead of entering marketing rings.

Prices Analysis and Pricing Policy

Organic fertilizers are sold to farmers at the agricultural products trading shops in plastic bags or bags (25) kg. Packages range from (1)JD for poor items to (2.5)JD for good and imported items. It is suggested that the price of (1) dinars for the bag with capacity of (25) kg. Thus, the price of ton is (40) JD of organic fertilizers, the price of the farm door to give a profit margin for the wholesaler and traders of agricultural materials to be sold to the farmer at a price ranging from (1.25 - 1.5) of the packaging capacity (25), and the project can obtain the remnants of agricultural crops by trade With farmers.

Market Share

The domestic market demand of organic fertilizers is covered by the use of poultry and livestock waste in its raw form in addition to the processed by some local companies as well as by importing from abroad in small quantities.

The project's market share of organic fertilizers has been estimated at about 4500 tons/ year by assuming that the plant will bring organic raw materials from farms in Ajloun and neighboring areas. The size of the project will determine the quantities of non-fermented fertilizers in Ajloun Governorate. Farmers will be contacted to obtain the solid residues from the crops, animal dung and poultry dung for fermentation, drying, sterilization and packaging to produce organic fertilizers. Table (9) shows the expected quantities of demand for organic fertilizers and the share of the project from market demand.

It has been assumed that the plant will produce organic fertilizers according to the instructions of the Ministry of Agriculture so that its production complies with the terms of organic fertilizers in Jordan according to the instructions No. G / 6 of 2016 of the terms and conditions of licensing the production of fertilizers and plant growth regulations and their processing, storage, circulation and trading. So that they are free of herbs seeds and exotic and odorless plants. As well as conforming to Jordan Standard Specification 962-2011 of fertilizers and organic soil fertilizers.

Table9 : expected quantities of demand for organic fertilizers and the share of the project

Year	Expected Demand (Tons)	Expected Production (Tons)	Market Share
2018	778,341	2250	0.3%
2019	795,464	4500	0.6%
2020	812,965	4500	0.6%
2021	830,850	4500	0.5%
2022	849,129	4500	0.5%
2023	867,809	4500	0.5%
2024	886,901	4500	0.5%
2025	906,413	4500	0.5%
2026	926,354	4500	0.5%
2027	946,734	4500	0.5%

Projected Revenues Analysis

Organic fertilizers can be produced throughout the year. Each fermentation batch requires two to two and a half months to complete maturity. It is assumed that ten organic fertilization cells will be set up to be operational throughout the year. , And for the purposes of this study it was considered that the average selling price is (40) JD/ ton of organic fertilizer and the prices of buying and selling for organic fertilizers is expected to increase by 2% per year.

Table 10: Selling prices and total expected sales during the period (2018-2027)

Revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Produced Quantity (Tons)	2,250	4,500	4,500	4,500	4,500	4,500	4,500	4,500	4,500	4,500
Prices (JD/Ton)	40	41	42	42	43	44	45	46	47	48
Total Revenues (Thousand JD)	90	184	187	191	195	199	203	207	211	215

4. Technical Analysis

Non-fermented organic fertilizer

When the non-fermented fertilizer (Animal dung) is added to the soil, the bacteria move towards it to ferment and perform the decomposition tasks, and to do that, we find that the bacteria consume the nitrogen to that they grow and multiply, then the competition starts between the plant and the bacteria on the soil nitrogen, which is definitely determined in favor of the bacteria. From this stage the bacteria begin to decompose the organic fertilizer and produce nitrogen and that time the plant has passed the stage of vegetative formation where the need is acute for nitrogen.

The use of fertilizer from the barn to the fields directly without going through the stages of decomposition outside the agricultural fields has lots of the damages limited to the following:

- 1- Extensive spread of exotic weeds by re-germinating their seeds in animal manure.
- 2- Nutrients are not readily available to be absorbed by the plant as this fertilizer takes a long time to pass through the decomposition stages.
- 3- This fertilizer is not decomposed. It is decomposed in the field of this plant by the bacteria, which compete with the plant on the nitrogen of these soil and the bacteria may damage the plant when it feeds on its roots.
- 4- The spread of harmful worms (nematodes), for example, and the presence of eggs and larvae of active insects.
- 5- The spread of flies in agricultural fields.
- 6- spread of roasted root diseases.

Therefore, animal and plant residues must be fermented outside the agricultural field to produce compost (to complete the process of decomposition and fermentation, so that the nutrients are easy to be add to the plant to benefit directly from them, and eliminate the seeds of weeds and harmful fungi and pathogens due to the arrival of temperature between (50 - 70) within the heap during the decomposition period at fermentation to produce the compost.

Fermented Organic Fertilizer (Compost)

Is a Latin term that means a mixture or a collection of multiple or different sources. Compost is the result of the biological decomposition of the organic material whether it is of plant or animal origin by bacteria and some useful microorganisms under certain environmental conditions of warmth, humidity and good ventilation. There are two types of compost according to the system and its production method (Pneumatic compost and non-pneumatic compost).

Importance and benefits of compost

The compost has a great importance to the plant through feeding the soil, it is considered the main key in the success of any agricultural project and also within it the spirit of life and growth, when it follows the optimal method of production, where we summarize and clarify that importance in the following:

- 1- Improves the properties of the soil and increases the preservation of moisture in it. When the organic material is dissolved in the compost and when the humus is formed, it leads to a complex formation of mud and humus, which helps porosity and sedimentation of the mixed and muddy soil and increases the cohesion of the loose or sandy soil which increase the preservation of Loss by evaporation or leakage away from the plant root area and also helps the possibility of washing salts when irrigation with salt water or agriculture in salt soils, as it has a role in increasing the fertility of calcareous soils
- 2- It prevents soil erosion by erosion factors such as water and wind, which erodes the roots and the corrosion of the fertile and rich of millions of useful microorganisms surface .
- 3- Provides the appropriate cradle for seed germination and root growth and diffusion.
- 4- Soil fertility is cumulative in contrast to chemical fertilization, which acts on toxicity and soil degradation cumulatively. Studies of mineral element show that 40% nitrogen, phosphorus and potassium are facilitated during the first year and 30% in the second and 30% The plant is available in the third year.
- 5- Helps to reach the desired pH value
- 6- To enrich the soil with beneficial microorganisms, it leads to the growth of millions of microorganisms beneficial of bacteria and aerobic fungi (the earthworm is present in the organic environment free of chemicals where it performs the decomposition of organic material and improves the properties of physical soil) Nitrogen air helps in the process of oxidation and release of phosphorus and potassium.

- 7- Soil purification, as a result of the activity of millions of beneficial microorganisms during the process of the camper and the production of compost, it consists of many of the hormones and enzymes and growth and natural stimulants that reach the soil, eliminate the fungus and pathogenic bacteria and improve the growth of the plant and have strong resistance
- 8- The disposal of toxic substances in soil, where good compost works on the mixture of harmful elements in the soil such as lead and cadmium by forming an association with the humus complex, making them unsustainable through the root of the plant.
- 9- The supply of major and minor elements to the plant, it provides major nutrients, such as nitrogen, phosphorus and potassium, in organic form and it is a good source to supply plants with micro-elements such as iron, manganese, copper and zinc.
- 10- The supply of plant nutrients in a systematic and gradual manner, to meet the nutritional need of the plant as a result of microbial activity in the soil, as these microbes feed on the organic matter and therefore the nutrients in the middle of the soil ready and easy to absorb the plant to them easily and easily, the content of this organic product safe and healthy and The storage time of this product increases in contrast to inorganic products.

Sources of agricultural waste and important food waste in the composition of compost:

There are many waste that can be recycled and used in several areas, the most important of which is the production of organic fertilizer, including:

- 1- Animal waste of all types of sheep and cows, especially animal manure.
- 2- Residues of all types of domestic birds and barn mattresses.
- 3- Residues of palm leaves, carbs, fibers and fallen or damaged fruits.
- 4- Residues of vegetables resulting from open fields or residues of green houses or damaged fruit.
- 5- The outputs of regular service work throughout the agricultural season, such as weeding, pruning and field cleaning results.
- 6- Falling leaves and pruning products of orchards, impurities and fruits resulting from sorting and grading of fruits.
- 7- Residues of windbreaks and perennial trees from fallen branches and leaves.

Organic fertilizer immersion systems

Anaerobic organic fertilizer

It is produced by an anaerobic process under ground or by a special mixer and is tightly covered with a non-permeable plastic cover to ensure that there is no air or oxygen in the compost heap with high humidity by immersing the pile with water. Moisture increased above than 60%, thus, the air is replaced by water.

This is the traditional method used by ancestral farmers. The degradation or fermentation of the anaerobic residues organic matter, plant or animal, is done in the absence of oxygen. The beneficial air bacteria suffocate and die and instead the anaerobic bacteria become active which in turn destroy the organic material but slowly and the resulting heat is insufficient to eliminate the causes. Where the oxidation of these substances is incomplete, leading to the formation and accumulation of organic acids and alcohol and noting the release of methane gas and hydrogen gas as well as sulfuric acid, which is responsible for bad smell.

During the manufacturing process we note that the length of manure manufacturing in this way need more time than the Aerobic method due to the thermal energy released in aerobic conditions for decomposition and fermentation processes is higher many times than the thermal energy levels produced in fermentation conditions or anaerobic degradation, so the final product is poor and smells foul and harms the plant.

Aerobics organic fertilizer

Aerobic process is the best way to produce high-quality organic fertilizers. This method is considered as one of the bioremediation methods for organic waste, whether of plant or animal origin, by beneficial bacteria that dissolves these materials when the appropriate environment of moisture is available (60%) and optimum temperature (60-70 degrees Celsius) we get a good organic fertilizer, A bacterial solution or farm may be added as a stimulant that helps in the rapid decomposition of these organic wastes. Its content of the humus increases, which works on enriching the soil with living organisms when added to it, stabilizing air nitrogen and dissolving phosphorus and potassium so that it can be easily absorbed by the plant.

Production Process

The dung is placed in a hole or U-shaped cement cells in several layers so that the first layer of manure is added and moistened a bit if it is dry and left for five days to allow room for aerial fermentation, and it is recommended to stir and move the manure after two days to ensure oxygen supply to the mass, The temperature has risen to 60 - 70 C and these conditions can kill harmful microorganisms as well as killing insect eggs and grass seeds, and after five days, a layer of 5-10 cm of hay or soil or plant leaves or a mixture is added and the mass is melded so that the conditions become ripe for the initiation of anaerobic fermentation and in these Stage temperature drops to 30-35 C. The second layer is added and treated in the same way. The process is repeated until the hole is filled. It is left for two months to obtain a half-fermented fertilizer or four months to obtain a fully fermented fertilizer, preferably the first bottom layer as well as the last top of the straw or soil or mixture of the two Together to absorb leaking liquids and volatile gases resulting from fermentation / decomposition. These drillings are done at a capacity that is commensurate with the needs of the project and according to the quantity of raw material available to it.

Terms and specifications of organic fertilizers

Organic Fertilizers: Fertilizers manufactured from plant or animal residues or mixtures thereof, including:

The following specifications must be met:

- The fertilizer should be fully decomposed and thermally treated
- Do not be mixed with soil or sand.
- free of weed seeds, viruses, bacteria, fungi, nematodes, pathogens, insects and animals.
- The content of toxic heavy metals does not exceed the permissible limit according to the following table:

Table11 : Fertilizer Contents

Element	Permissible limit (ppm)	Element	Permissible limit (ppm)
Larsen	15	Lead	120
Cadmium	3	Mercury	1.5
Crome	100	Nickel	50
Cupper	150	Zinc	350
Selenium	4	mulbdenum	18

- The percentage of organic material in locally grown fertilizers should not be less than 40%
- The electrical conductivity shall not exceed 10 dS / m and the pH shall be 7.5 in 5: 1 (organic dry weight: water)
- The humidity should not exceed 25%.
- The ratio of carbon to nitrogen should not exceed 1:20 (C / N)
- Sodium chloride in locally produced and imported fertilizers shall not exceed 2% and that the dissolved sodium should not exceed 0.08%
- The radiation should not exceed 300 piecor / kg.
- The weight of the compost package should be 25 or 50 kg and shall be closed, solid and bear the circulation.
- To indicate on the packaging the type of fertilizer, trade name and other information related to production.

Raw Materials

Many substances are needed to produce organic fertilizer because organic fertilizer is a mixture of organic material from various sources. The following tables show the cost of raw materials to produce one ton of organic fertilizers.

Table 12: Required materials to produce one ton of fermented organic fertilizer

Production Requirement	Required Quantity	Unit Price	Total Cost
Unit	(KG/Ton)	(JD)	Ton Cost
Agricultural waste	1,000	2.5	2.5
Cattle Dung	250	4	1.0
Poultry manure	250	4	1.0
Leguminous or green residues	100	4	0.4
Agricultural soil	50	4	0.2
Mature Compost	50	1	0.1
Urea	5	400	2.0
Phosphate	20	50	1.0
Feldspar	20	50	1.0
Dolomite	5	50	0.3
Agricultural sulfur	5	100	0.5
Plastic bags for packing	40	0.06	2

Production Requirement	Required Quantity	Unit Price	Total Cost
Unit	(KG/Ton)	(JD)	Ton Cost
Plastic Covers	1	0.10	0.10
Water	0.20	0.50	0.10
Total	1,755		12.5

The main material is the residues of poultry and livestock farms, which are found in Jordan, and some fertilizer manufacturers obtain this material from the farms for free versus the continuous collection of those wastes and cleaning the farms from them throughout the year, while others buy them at a price ranging between 3 Dinars and 5 dinars per ton, according to the availability of the material and as noted earlier, some livestock breeders get rid of the manure in different ways to prevent tamping in the landfill of waste or burning or throw it out,

As well as 25-50 kg bags are required, they can be obtained from the local factories at 60 fils per bag, in addition to hay and mixing with the waste during fermentation at low rates.

The investor can get most of his raw material needs free of charge, especially animal dung and poultry, and buy the rest at the above prices. The above prices include purchase of production inputs, in addition to the prices of transport costs to the project site. It is estimated that the cost of raw materials per ton is about 12.5 dinars. Each ton of ready fertilizer needs about 1.65 tons of raw materials, with a return rate of raw materials of 65%. Animal residues usually contain a high percentage of moisture and some impurities, the cost of packing materials per ton of ready-made fertilizers will be about 2JD.

Required Human Resources

The production of compost requires unskilled workers working one shift daily, in addition to an agricultural engineer specialized in organic fertilizer. One person is sufficient to manage the 10,000 m² unit, which is the place of fermentation, cleaning and sterilization organic fertilizers. Assuming that the owner may devote his time to make deals with farmers and to negotiate the purchase of production inputs of crop residues and animal dung, and perhaps he needs a second person (production supervisor) when it comes to bringing production materials or loading manure from farms and crop residues from fields, The project also needs two drivers, one for the dump and one for the bulldozer.

The following table shows the employment requirements:

Table 13: Required Manpower

Job Title	No.
Indirect Employees	
Project Manager / Plant Production Engineer	1
Production Supervisor	1
Total Indirect Employees	2
Direct Employees	
Cultivation workers	4
Driver	2
Total Direct Employees	6
Total Employees	8

Job Description:

- Agricultural Engineer:** A full knowledge of how organic fertilizers are manufactured, how they can be processed, and the ability to request and provide the necessary materials from production tools such as manure and crop residue, and the implementation of prevention measures and pest control in the product in accordance with environmental standards. Detection of organic manure during fermentation process and know the maturity time. Supervising the flipping of compost during the preparation stage. Preparation and packaging, as well as the filling of farm records and business models, documenting expenditures, proposing policies and action plans, monitoring their implementation and evaluating their results. Organizing daily work, distributing workers, and following up on their work, in coordination with farmers, suppliers and transport companies to ensure the safety of the supply of raw materials and production.
- Production supervisor:** inspect the equipment and tools of the organic fertilizer production unit, such as: fermentation tanks and solid waste filling tools and equipment, such as: bulldozers, conveyor belts, screening equipment and packing. Preparing the organic fertilizer production plant and equipment, operating, and monitoring according to manufacturer's instructions. Placing the solid waste in the fermentation tanks, adjusting the conditions of the fermentation process such as temperature and humidity, and monitoring the fermentation process according to the instructions. Operation of organic manure extraction equipment, extraction of compost from holes, operation of

screening equipment and packing of organic fertilizer produced according to instructions. Service of organic manure production equipment and tools and cleaning the compost, also cleaning the workplace. Record readings of temperature and humidity of measuring instruments and organic fertilizer product quantities. Application procedures and instructions of the Occupational Safety and Health.

- **Agricultural worker:** Works under the supervision of the agricultural engineer and prepares the necessary materials for the process of fermentation, sterilization, processing and packaging and the application of procedures and instructions for occupational and public safety and health.
- **Product distribution driver:** Operates under the supervision of the engineer and brings raw materials and distribution of the product to the places of marketing, applying the procedures and instructions of occupational and public health and safety, driving agricultural trailers and machinery, the operation of the rear rotate column and the suspension and semi-hanging machines and the identification of types of mechanical couplings and places of use. The operation of soil preparation machines attached to the tractor of different types, and the use of each type of them according to the purpose of use, and to make adjustments and service.

Site Analysis

It is recommended that the project is to be set up in one of the areas where there are farms for livestock and poultry and winter and summer crops for easy access to the raw material (manure) and transfer them to the plant easily and at the lowest cost. The plant needs a large plot of land of not less than 10 dunums, where the bulk of it is used for manure and manure after it is collected from the farms, preferably the plant site away from population centers and water sources to prevent pollution. .

When selecting a project site, the following should be considered:

- Easy marketing and transportation of raw materials and processed products.
- Provision of a sustainable source of water for agriculture.
- Far from residential neighborhoods.
- All necessary services of paved roads, water and electricity should be available In the proposed area.

Based on all these data, it is proposed to establish in Ajloun a plant for the production of organic fertilizers, so that the appropriate location for production is chosen based on the availability of infrastructure services of water and electricity and the availability of raw materials. For example, Away from population centers and land prices are low. The project should be away from the housing because of the smells that may escalate. Should be in the center of nature, or within the fields or close to cultivation.

Description of the production process and Manufacturing steps

Basic stages of production of compost

There are three main characteristics of fermented manure during the decomposition period, which occur from the beginning of the preparation until the maturity and decomposition of the contents of this fertilizer.

- 1- Heating stage
- 2- Cooling stage
- 3- Stage of perfusion

Table 14: The basic stages of production of fermented organic fertilizers (compost)

Stage	Temperature	Organisms	Work	Products
Heating stage (Thermophiles)	(20-50) degrees Celsius	Bacteria and fungi that live at a moderate temperature	Fragmentation of carbohydrate proteins and starch	Ammonia, nitrate, carbon dioxide, polysaccharides, water and minerals
	(50-70) degrees Celsius	Microorganisms	Extra fragmentation of complex compounds	
Refrigeration phase	(25-50) degrees Celsius	Microorganisms live at a moderate temperature	Dissolution of cellulose and lignin	Ammonium nitrate, carbon dioxide, water,
Decomposition stage (Cycroflex)	(20-25) degrees Celsius	Microorganisms live at a moderate temperature	Combination of mineral nutrients	Humus and more solid elements
		Soil animals (such as worms and spiders)	Mixing organic and mineral compounds	Fermented fertilizers (Decomposed)

Source: Fermented Fertilizer Production Manual. Project of Organic Agriculture Development .
Ministry of Agriculture, Saudi Arabia 2007 (Series of Organic Agriculture Publications).

Aerobics immersion operation is a vital process that depends on the metabolic activity of many microorganisms. These organisms depend on the nutrients of these plant and animal residues. These microorganisms, such as bacteria and fungi, begin to break down carbohydrates, nitrogen and starch. Large amounts of heat and carbon dioxide And water vapor.

As a result of this heat, the convection temperature is 60 ° C, which helps eliminate the weed seed embryos and also eliminate pathogens such as microbes, fungi and nematodes. In addition, these microorganisms secrete many antibiotics that help to eliminate pathogens during the decomposition period or when they are added to the soil, where they have a major role in the protection of the plant through the resistance and increase the immunity and therefore the need to use pesticides. In addition, microorganisms secrete some natural growth regulators and growth factors that improve crop characteristics.

The microbial or biological compost is the product of the process of decomposition by microorganisms which is controlled and directed to obtain the desired product by continuous ventilation by the flipping machine attached to tillage, or by hand. The optimal conditions for the microorganisms that are added to the compound materials are provided through the activated solution Bacterial or organic manure during the process of demolition of organic material decomposition, construction of humus and resulting materials and maintaining of aerobic conditions.

It is preferable to add clay soil or bentonite (clay) to the composting process by (5%) of the raw material involved in the process. The microorganisms need to have clay minerals during their decomposition activity to help build humus and clay. Links are made between clay metal and organic component.

- It is also recommended to add (5-10%) of the mature compost as it contains the most important microorganisms necessary for the process, as well as help in improving the humidity and ventilation within the new production line at the beginning of the preparation process and the immersion.

To satisfy the nutritional needs of plants from phosphorus, potassium and magnesium, natural sources such as powders of mineral ores can be used from natural rock (5-10 kg / ton).

Production of fermented organic fertilizer

Organic residues are collected from several sources in a central and specific location called the fermented manure production unit. This method is characterized by placing a place that has the required specifications, with all the possibilities and supporting needs and assistance in producing it.

- 1- Low-level land should be avoided to avoid pooling during rainfall and avoid rising ground water as this reduces the efficiency of decomposition within the heap.
- 2- The choice of an area of integrated flat land preferably to be a concrete slope ends in drains through which the leakage of leachate is collected and returned again to the matrix as well as ensure that the fertilizer is not mixed with rocks and earth soil in addition to the ease of movement of the equipment during the fertilizer service.
- 3- The production site of the compost should be close to the place of production of the farm waste and next to the machines of shredding those wastes to reduce the cost of transportation.
- 4- The compost production site should be close to the water source or otherwise water source should be provided..
- 5- The location of the compost production unit is shaded so that the arrays are protected from vertical sunlight, especially in the summer, which works on the release of nutrients, as well as to avoid the washing processes of these elements during precipitation. The compost unit must be open to ensure access to beneficial side sun rays and Air from all directions due to its importance during the period of aerobic decomposition.
- 6- The matrices are built hierarchically, with a base width of 2 m to 2.5 m and height of 1.5 m to 2 m depending on the softness and roughness of the organic components of the pile or matrix, and free length according to the available place of the farm.

Table 15: Raw materials of compost

Type of organic waste	Sources of these wastes
Dried brownish plant residues. (Slow-degradable materials, concentrated in lignin and cellulose and containing high carbon content)	Palm leaves, fibers and all dry wood residues are also used to produce hay, as well as non-colored newspaper leaves and office papers
Green plant residues (rapid decomposition containing high	Crop residues of all kinds, weeding products, harvest of green fields, tree

Type of organic waste	Sources of these wastes
nitrogen content)	trimming, green leaves, falling fruit, sorting and grading products of fruits, vegetables, and the resulting impurities during sorting and grading
Residues of animal production (Nitrogenous)	Dung, mattress, brushes, horns, blood and Bone powder
Poultry residues (nitrogenous)	Feathers, White Peels and manure
Compost Organic Fertilizer	5-10% of the amount of input mixture
Agricultural clay soil or bentonite metal	5-10% of the amount of input mixture
Some natural fortifiers and rocks. Compound and mixture of natural mineral ores and ores that are added to the fermented manure production line at maturity and after the completion of the aerobics immersion process	— Phosphate rock as a source of phosphorus. Feldspar rock as a source of potassium. Magnetic iron oxide. Agricultural sulfur. Dolomite rocks as a source of calcium and magnesium.

There are some undesirable substances that require the removal before the use of these agricultural or animal plant residues. Some materials related to unwanted agricultural waste:

- 1- Plastic, glass, metal, rubber and steel parts.
- 2- Chemicals, colored Journals and wood Painted Fabrics.
- 3- Meat, grease, oils, leather and dead animals.

A Compost Matrix Building Method

The matrix is constructed in a sequential manner for the layers of various organic materials, which are constructed with the following steps:

- 1- Flat land with a width of 2.2 x 5 meters and free length shall be determined according to the area available on the site.
- 2- The area allocated for the pile is chosen on the basis that the ton of compost needs 2 * 3 m 2.
- 3- And dig a channel to collect water, at an altitude of 10 cm, width of 20 cm and a depth of 10 cm ends with a pool to collect the leachate so that it can be reused in the spray again.
- 4- Spray the water on th selected floor to create the moisture underneath the heap before you start brushing the layers.

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- 5- A layer of heavy dry plant residues are placed to be porous allowing the air to permeate the bottom of the heap and then sprinkle with water.
- 6- Followed by a layer of animal residue (bovine manure) with a thickness of about 30 cm and sprayed with water.
- 7- Then a layer of light dry plant wastes, such as leaves or plants that have been chopped, with a thickness of 30 cm and sprinkled with water.
- 8- Then a layer of poultry residue with a thickness of 15 cm and sprinkle with water.
- 9- Followed by a layer of mature compost (thickness 15 cm) and then sprinkled with water.
- 10- Followed by a layer of light green vegetable leaves (preferably a bean) 30 cm thick and then sprinkled with water.

These layers are repeated until the height reaches 2.5 meters according to the softness and roughness of the organic components.

Finally, a layer of soil (5-10 cm) is sprayed with water (the goal of the final coverage of this soil is to build humus and prevent the spread of flies), and then sprinkle a layer of natural mineral rock powder upon maturity.

Using what is available in the area from animal and plant agricultural wastes and the diversity of organic sources is directly proportional to the quality of the compost content of the nutrients.

Periodic flipping of the matrix, spraying and periodic moisturizing with water (where the flipping machine is used), the optimum humidity shall be defined not to exceed 60% (either by measuring the moisture content by the measuring tool or by taking a sample of 40 cm depth inside the heap or matrix) Decomposition Bacterial fluid can be added during flipping during the preparation and mixing of wastes or when the temperature within the heap reaches 60-70% within the heap.

It is recommended that the flipping be repeated once a week during the decomposition period. This is done by following the temperature to no more than 70 C within the heap until the appearance of the compost maturation signs as a decrease in the temperature and moisture content and the degradation of the plant waste and its transformation into spongy, It is preferable to leave the compost pile for at least two weeks for the final maturity and to allow the chemical transformations and the composition of the humus complex. The compost is expected to reach full maturity after about 2.5-3 months And about 4 months in winter, from the beginning of the aerobics immersion of the heap.

Model of raw materials used to produce 1 ton of compost (1.5-2 m³) for use in organic cultivations. This model consists of two main parts:

1- The first part: materials are added at the beginning of composting

1 ton of Agricultural dry and chopped crop waste.

250 kg of cow dung

200 kg of poultry manure

100 kg of legume or green waste .

50 kg of agricultural soil or clay.

50 kg mature compost

The total weights of the raw materials of this group to 1650 kg

2- The second part: Substances are added after the immersion and before maturity.

20 kg phosphate rock as a source of phosphate.

20 kg Veldspar rock as a source of potassium.

5 kg dolomite rock as a source of calcium and magnesium.

5 kg agricultural sulfur and agricultural gypsum (as a source of sulfur).

As a result of biodegradation and aerobic immersion process loss occur in the weight and final production to about 65% of the weight of materials at the beginning of the immersion; 1650 kg raw materials give 1 ton of compost ranging between 1.5 - 2 cubic meter.

Signs of organic fertilizer maturity

Usually maturity is after 2.5-3 months in summer and may be up to 4 months in winter. This time period may not be fixed due to the influence of many factors on maturity, such as the quality of the raw organic material entering the compost and the size of the waste. And the temperature inside organic compost matrices through water spraying and periodic flipping of those matrices.

The mature manure should be stored until it is well pressed and reduced in size, protected from the heat of the sun and wind and covered with hay, cut from the burlap or any access cover, while keeping the water moistened with water. Or filled with bags after sifting and grinding to be ready for use on the farm or ready to market

Table 16: Specifications of mature fermented manure (good compost)

Characteristics	Description and optimal limits
Color	It can be dark brown or sepia
Texture	Spongy
Odor	The smell is acceptable as the smell of soil sprinkled with water
the weight	The weight of the cubic meter does not exceed 700 kg / m
Tempreture	The temperature of the outer atmosphere is 5-10%
Humidity	The moisture content shall not exceed 30%
PH	should not exceed 8
Oxygen	That not less than 5%
Carbon Dioxide	Shall be between 1- 2%
Total nitrogen	That not less than 1%
Nitrogen ammonia	Be between 50 -100 mg / kg
Nitrogen Nitrate	Be between 200 -300 mg / kg
Nitrite Nitrogen	None
Total phosphorus	That not less than 0.8%
Total potassium	That not less than 1%
Ash	To be between 70- 80%
Organic Material	Not less than 30%
Humus	Between 25-30%
Hydrogen sulfide	None
Carbon ratio of nitrogen	1:10

Technical Requirements Analysis

Process requirements for the production of organic fertilizers

- 1- The diversity of the materials involved in the aerobics immersion process and their variation to contain the residues of dry, green, leguminous, animal and poultry waste.
- 2- Waste sorting of plastic, metal, glass and colored paper.
- 3- Adjust the moisture content of the contaminants to maintain the warmth during the aerobics immersion period so that it does not exceed (50%) by the wet or primitive measuring device or by taking a sample of 40 cm depth within the matrix.
- 4- Periodic follow-up of the temperature by means of a temperature measuring device not exceeding (70) centigrade or by the method of examination and sensitivity by hand when taking a sample from the depth of 40 cm within the matrix or to observe the presence of traces in this sample.
- 5- (5) to control the size of the matrix so that the base width (2 to 2.5) meters and the height between (1.5-2) meters according to the components of organic material, the more compost components of organic material was rough and heavy, the more height required and the more soft and light materials, the height is lower and the length of the matrix is not determined by the place available on the farm.
- 6- Flipping once a week (or depending on the temperature and internal humidity of the matrix so that the temperature does not exceed 70 degrees Celsius and the humidity does not exceed 60%) and flipping is done with the appropriate way based on the amount of compost.
- 7- The flipping machine for the production of organic fertilizer, as it is a typical machine for the rotation of manure, it works on the ventilation and fragmentation of blocks and homogenization of compost components while ensuring that the manure is not mixed with soil and small stones during periodic flipping of the matrix.

Land and Construction Works

The compost project needs a piece of empty and flat land with an area of about 5000 square meters. It is used in fermentation works and a separate warehouse for storage, in addition to a small administrative building attached to a water cycle. The total area of construction is 250 m². Therefore it is proposed to establish the project on a plot of land area of 10,000 m², allocates 5000 square meters for fermentation cells and the rest of the area of movement and processing materials and the remaining space for services, facilities and aisles. The following table shows the required land.

Table17 : Required land

Land	Area (sqm)
The Required land	10,000

Table 18: Required construction works

Construction works	Area (sqm)
Management Offices and Housing	100
Warehouses	200
Fermentation Cells	5,000
Open Spaces and Garages	1,000
Bathrooms and Workers Restrooms	50

Machinery and Equipment

Equipment mainly consists of a tank to receive the raw material and mill that can be obtained from the local market where they can manufacture this line with a production capacity of about 40 tons per day, as the project will need to dump to transfer the raw material of poultry and livestock farms to the factory site and the transfer of ready-made goods to farms, nurseries and shops selling agricultural materials, as the project will need a bulldozer used in individual raw material in the land of the project and the drilling holes prepared for fermentation, transport operations as well as to transfer this material to the production line (reception of the raw material tank) and not necessarily to be new, a balance to weigh the final product when packaged in bags, two hand stitching machines for bags and the number of tools and sporadic.

The following table provides the machinery and equipment needed for the project:

Table 19: Required Machinery and Equipment

Machinery and Equipment	Quantity
Milling machine for plant waste	1
A machine for mixing agricultural waste	1
Sieve	1
Flipping Machine	1

Machinery and Equipment	Quantity
Screening and softening compost machine	1
water tank	2
Hand instruments for measuring the temperature, humidity and acidity.	1
Packing and packaging unit	1

Furniture

The project needs the following furniture:

Table 20: Required Furniture

Furniture	Quantity
Offices Furniture	1

Vehicles and Transportation

The project needs the following Vehicles and transportation means:

Table 21: Required Vehicles and Transportation

Vehicles	Quantity
A large transport vehicle	1
Loder	1

5. Financial Analysis

Assumptions

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 0 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Table 22: General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 23: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 24: Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 25: Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	0.00%
Utilities Expenses from Total Revenues	2.00%
Maintenance Expenses from Total Revenues	0.05%
Depreciation Expenses from Total Revenues	0.05%
Insurance Expenses from Total Assets Value	0.00%
Marketing Expenses from Total Revenues	1.00%

Table 26: Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	100%
Income Tax after Deduction	0%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	Value
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Table 27: Risk Premium

Risk Premium	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	7.50%
Market Risk Premium (3)	9.93%

Income Tax Rate	0.00%
Beta (4)	1
Growth rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 28: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premium	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	128,992
Loans Value	128,992
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50
Loans Value	128,992
Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	128,992
Equity Percentage	%50
Gross	
Project Value	257,984
Proportion of the project	%100

Capital Expenditures

The estimated cost of the project is 234,531 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 29: Land Capital Expenditure

Land	Cost) Jordan Dinar/Meters Square(	Area (M2)	Cost
Required Land	5	10,000	50,000
Total			50,000

Table 30: Construction Work Capital Expenditure

Description	Cost (JOD per Squared	Area	Gross Cost
Management Offices and Housing	50	100	5,000
Warehouses	80	200	16,000
Fermentation Cells	6	5,000	30,000
Open Spaces and Garages	5	1,000	5,000
Bathrooms and Workers Restrooms	50	50	2,500
Others	-	-	5,850
Gross Total			64,350

Table 31: Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Milling machine for plant waste	1	2,000	2,000
A machine for mixing agricultural waste	1	3,000	3,000
Sieve	1	1,000	1,000
Flipping Machine	1	10,000	10,000
Screening and softening compost machine	1	8,000	8,000
water tank	2	1,000	2,000
Hand instruments for measuring the temperature, humidity and acidity.	1	10,000	10,000
Packing and packaging unit	1	500	500
Gross Total			36,500

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Table 32: Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Office Furniture	1	1,000	1,000
Gross Total			1,000

Table 33: Transportation and Vehicles Capital Expenditure

Transportation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
A large transport vehicle	1	18,000	18,000
Loder	1	18,000	18,000
Gross Total	2		36,000

Table 34: Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	500
Transportation	500
Marketing	1,000
Training	500
Total	2,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 35: Initial Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	6	%50	29,024
General and administrative expenses	3	%25	213
Marketing expenses	3	%25	145
Indirect Salaries	3	%25	4,800
Total			34,181

Table 36: Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	60,000	10%	66,000	%25.6
Construction works	64,350	10%	70,785	%27.4
Machinery & Equipment	36,500	10%	40,150	%15.6
Furniture	1,000	10%	1,100	%0.4
Vehicles	36,000	10%	39,600	%15.3
Pre-Operating Expenses	2,500	10%	2,750	%1.1
Working capital	34,181	10%	37,599	%14.6
Total (JD)	234,531		257,984	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (257,984) JD, where 50% of the project will be financed through loans (128,992) JD, and the remaining (50%) through (Equity) with (128,992) JD. The following table summarizes the general structure of the required financing:

Table 37: Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	128,992	%50
Loans	128,992	%50
Total	257,984	%100
Use of Fund	Value	Ratio
Capital Expenditures	217,635	%84.36
Pre-Operating Expenses	2,750	%1.07
Working Capital	37,599	%14.57
Total	257,984	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 38: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	28,125	56,250	56,250	56,250	56,250	56,250	56,250	56,250	56,250	56,250
Utilities from Revenues	1,800	3,672	3,745	3,820	3,897	3,975	4,054	4,135	4,218	4,302
Maintenance from Revenues	0	92	94	96	97	99	101	103	105	108
Disposables from Revenues	45	92	94	96	97	99	101	103	105	108
Direct Salaries	22,800	31,312	32,251	33,219	34,215	35,242	36,299	37,388	38,510	39,665
Others	5,277	9,142	9,243	9,348	9,456	9,567	9,681	9,798	9,919	10,043
Total	58,047	100,559	101,677	102,828	104,013	105,232	106,487	107,778	109,108	110,476

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 39: General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	200	204	208	212	216	221	225	230	234	239
Hospitality	100	102	104	106	108	110	113	115	117	120
Stationary and Printing	50	51	52	53	54	55	56	57	59	60
Miscellaneous	500	36	36	37	38	39	39	40	41	42
Total	850	393	401	409	417	425	434	442	451	460

Marketing Expenses

Table 40: Marketing Expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	580	1,006	1,017	1,028	1,040	1,052	1,065	1,078	1,091	1,105

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 41: Expected monthly salaries (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project Manager / Plant Production Engineer	700	721	743	765	788	811	836	861	887	913
Production Supervisor	500	515	530	546	563	580	597	615	633	652
Direct Employees										
Cultivation workers	300	309	318	328	338	348	358	369	380	391
Driver	350	361	371	382	394	406	418	430	443	457

Table 42: Expected yearly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project Manager / Plant Production Engineer	11,200	11,536	11,882	12,239	12,606	12,984	13,373	13,775	14,188	14,613
Production Supervisor	8,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Total Indirect Salaries	19,200	19,776	20,369	20,980	21,610	22,258	22,926	23,614	24,322	25,052
Direct Employees										
Cultivation workers	14,400	19,776	20,369	20,980	21,610	22,258	22,926	23,614	24,322	25,052
Driver	8,400	11,536	11,882	12,239	12,606	12,984	13,373	13,775	14,188	14,613
Total Direct Salaries	22,800	31,312	32,251	33,219	34,215	35,242	36,299	37,388	38,510	39,665
Total Salaries	42,000	51,088	52,621	54,199	55,825	57,500	59,225	61,002	62,832	64,717

Depreciations

:The following tables show all equipment costs, expenses and depreciation rates

Table 43: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	66,000	%0.0	%0.0
Construction Works	70,785	%5.0	%0.0
Machinaries	40,150	%10.0	%2.0
Furniture	1,100	%10.0	%5.0
Vehicles	39,600	%10.0	%0.0
	217,635		

Table 44: Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	217,635	218,493	219,370	220,266	221,182	222,118	223,075	224,053	225,052	226,074
Total Depreciation Expenses	11,624	11,710	11,798	11,887	11,979	12,073	12,168	12,266	12,366	12,468
Total Accumulated Depreciation	11,624	23,334	35,132	47,019	58,998	71,071	83,239	95,505	107,871	120,339
Total Additions	0	858	877	896	916	936	957	978	1,000	1,022
Total Net Book Values	206,011	195,159	184,238	173,247	162,183	151,047	139,836	128,548	117,181	105,735

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 45: Loan Details

Loan Value	128,992
Annual Interest Rate	9.50%
Loans Period	5
Loan Starts at year:	0
Annual Payment	2018

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2017				128,992
2018	33,594	12,254	21,340	107,652
2019	33,594	10,227	23,367	84,285
2020	33,594	8,007	25,587	58,698
2021	33,594	5,576	28,018	30,680
2022	33,594	2,915	30,680	0

Income Statement

Table 46: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	90,000	183,600	187,272	191,017	194,838	198,735	202,709	206,763	210,899	215,117
Gross Operating Revenues	90,000	183,600	187,272	191,017	194,838	198,735	202,709	206,763	210,899	215,117
Operating Expenses	(58,047)	(100,559)	(101,677)	(102,828)	(104,013)	(105,232)	(106,487)	(107,778)	(109,108)	(110,476)
Gross Operating Profit	31,953	83,041	85,595	88,189	90,825	93,503	96,223	98,985	101,791	104,641
Gross Profit Percentage	%36	%45	%46	%46	%47	%47	%47	%48	%48	%49
Salaries and Benefits (Indirect Staff)	(19,200)	(19,776)	(20,369)	(20,980)	(21,610)	(22,258)	(22,926)	(23,614)	(24,322)	(25,052)
General and Administraive Expenses	(850)	(393)	(401)	(409)	(417)	(425)	(434)	(442)	(451)	(460)
Markeing Expenses	(580)	(1,006)	(1,017)	(1,028)	(1,040)	(1,052)	(1,065)	(1,078)	(1,091)	(1,105)
Pre Operating Expenses	(2,750)									
Gross Indirect Expenses	(23,380)	(21,174)	(21,787)	(22,417)	(23,067)	(23,735)	(24,424)	(25,134)	(25,864)	(26,617)
Income before Interest, Depreciation and Tax	8,573	61,866	63,808	65,772	67,758	69,767	71,798	73,852	75,927	78,024
Fixed Assets Depreciations	(11,624)	(11,710)	(11,798)	(11,887)	(11,979)	(12,073)	(12,168)	(12,266)	(12,366)	(12,468)
Income before Tax and Interests	(3,052)	50,156	52,010	53,885	55,779	57,695	59,630	61,586	63,561	65,556
Bank Interests	(12,254)	(10,227)	(8,007)	(5,576)	(2,915)	0	0	0	0	0
Income Before Tax	(15,306)	39,929	44,003	48,308	52,865	57,695	59,630	61,586	63,561	65,556
Income Tax	0	0	0	0	0	0	0	0	0	0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	(15,306)	39,929	44,003	48,308	52,865	57,695	59,630	61,586	63,561	65,556
Net Profit Percentage	%17-	%22	%23	%25	%27	%29	%29	%30	%30	%30
Compulsory Reserves	0	(3,993)	(4,400)	(4,831)	(5,286)	(5,769)	(5,963)	(6,159)	(6,356)	(6,556)
Retained Earnings	(15,306)	20,630	60,233	103,711	151,289	203,214	256,881	312,308	369,513	428,514

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 47: Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	(15,306)	39,929	44,003	48,308	52,865	57,695	59,630	61,586	63,561	65,556
Non-monetary adjustments										
Bank benefits	12,254	10,227	8,007	5,576	2,915	(0)	(0)	(0)	(0)	(0)
Consumption	11,624	11,710	11,798	11,887	11,979	12,073	12,168	12,266	12,366	12,468
Operating cash flows before changes in working capital	8,573	61,866	63,808	65,772	67,758	69,767	71,798	73,852	75,927	78,024
Shortage (increase) in inventory	(2,344)	(2,344)	0	0	0	0	0	0	0	0
Decrease (increase) in accounts receivable	(7,500)	(7,800)	(306)	(312)	(318)	(325)	(331)	(338)	(345)	(351)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
(Decrease) increase in accounts payable	2,494	1,199	93	96	99	102	105	108	111	114
Change in working capital	(7,350)	(8,945)	(213)	(216)	(220)	(223)	(227)	(230)	(234)	(237)
Net cash flows (used) from operating activities	1,222	52,922	63,595	65,556	67,539	69,544	71,572	73,621	75,693	77,787
Cash flows from investing activities										
(Purchase) of fixed assets	(217,635)	(858)	(877)	(896)	(916)	(936)	(957)	(978)	(1,000)	(1,022)
Net flows (used) from investment activities	(217,635)	(858)	(877)	(896)	(916)	(936)	(957)	(978)	(1,000)	(1,022)
Cash flows from financing activities										
Equity capital	128,992									
Loan Amortization	(21,340)	(23,367)	(25,587)	(28,018)	(30,680)	(0)	(0)	(0)	(0)	(0)
Bank Interest Rate	(12,254)	(10,227)	(8,007)	(5,576)	(2,915)	0	0	0	0	0
Loans	128,992	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	224,390	(33,594)	(33,594)	(33,594)	(33,594)	0	0	0	0	0
Net increase (decrease) in cash	7,977	18,469	29,124	31,065	33,029	68,608	70,615	72,643	74,693	76,765
Cashflows at the Beginning of Period	0	7,977	26,447	55,571	86,636	119,665	188,273	258,888	331,531	406,224
Cashflows at the End of Period	7,977	26,447	55,571	86,636	119,665	188,273	258,888	331,531	406,224	482,989

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 48: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	7,977	26,447	55,571	86,636	119,665	188,273	258,888	331,531	406,224	482,989
Inventory	2,344	4,688	4,688	4,688	4,688	4,688	4,688	4,688	4,688	4,688
accounts receivable	7,500	15,300	15,606	15,918	16,236	16,561	16,892	17,230	17,575	17,926
Total assets traded	17,821	46,434	75,864	107,242	140,589	209,521	280,467	353,449	428,487	505,603
Non-current assets										
Fixed assets (net)	206,011	195,159	184,238	173,247	162,183	151,047	139,836	128,548	117,181	105,735
Total non-current assets	206,011	195,159	184,238	173,247	162,183	151,047	139,836	128,548	117,181	105,735
Total assets	223,832	241,593	260,102	280,488	302,772	360,568	420,303	481,996	545,668	611,338
Current liabilities										
Accounts Payables	2,494	3,692	3,786	3,882	3,980	4,082	4,186	4,294	4,405	4,519
Remaining amount of Loan	23,367	25,587	28,018	30,680	0	0	0	0	0	0
Total current liabilities	25,861	29,280	31,804	34,561	3,980	4,082	4,186	4,294	4,405	4,519

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-current liabilities										
long term loans	84,285	58,698	30,680	(0)	(0)	(0)	(0)	(0)	(0)	(0)
Total long - term liabilities	84,285	58,698	30,680	(0)	(0)	(0)	(0)	(0)	(0)	(0)
Total liabilities	110,146	87,977	62,483	34,561	3,980	4,082	4,186	4,294	4,405	4,519
Property rights										
Members Contributions	128,992	128,992	128,992	128,992	128,992	128,992	128,992	128,992	128,992	128,992
Compulsory reserve	0	3,993	8,393	13,224	18,511	24,280	30,243	36,402	42,758	49,313
Profit (loss)	(15,306)	20,630	60,233	103,711	151,289	203,214	256,881	312,308	369,513	428,514
Total equity	113,686	153,615	197,619	245,927	298,792	356,487	416,117	477,702	541,263	606,819
Total liabilities and equity	223,832	241,593	260,102	280,488	302,772	360,568	420,303	481,996	545,668	611,338

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 49: Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	(257,984)	3,972	52,064	62,718	64,660	66,623	68,608	70,615	72,643	74,693	76,765	1,002,327
Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash Flows	(257,984)	3,548	41,531	44,684	41,144	37,863	34,824	32,013	29,413	27,011	24,794	323,733

Table below (50) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 50: payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(257,984)	3,972	52,064	62,718	64,660	66,623	68,608	70,615	72,643	74,693	76,765
Project Value Returned	257,984	254,012	201,948	139,230	74,570	7,948	(60,660)	(131,275)	(203,919)	(278,612)	(355,377)
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 51: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.97
Net Present Value for Cashflows	382,571
Payback Period	5
Internal Rate of Return	%25.59

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 52: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	25.59%	5	11.97%
Revenues declined by 10%	19.40%	6	11.97%
Operating Expenses Increased by 10%	19.61%	6	11.97%
Total Expenses Increased by 10%	18.14%	7	11.97%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 382,571 Jordan Dinars considering that the project provides 8 Job Opportunities for the governorate residents.